BIOGRAPHICAL SKETCH

Marcus DuPont
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(a) Education & Training

New York University	New York, NY	Physics	Ph.D., 2024
	Advisor: Andrew MacFadyen		
	Exploring Novel Mechanisms Behind Astrophysical Transients		
New York University	New York, NY	Physics	MPhil., 2023
Florida State University	Tallahassee, FL	Physics and Astrophysics	B.S., 2019

(b) Research & Professional Experience

Postdoctoral Fellow, Princeton University	Sep 2024 – Present
Data Science Fellow, LSST-DA Data Science Fellowship Program	Sep 2022 – Jun 2024
Research Fellow, Max Planck Institute for Astrophysics	Jun 2023 – Aug 2023
Research Fellow, Center for Astrophysics | Harvard & Smithsonian	Jun 2019 – Aug 2019
Research Fellow, Center for Astrophysics | Harvard & Smithsonian	Jun 2018 – Aug 2018
Research Fellow, Center for Astrophysics | Harvard & Smithsonian	Jun 2017 – Aug 2017

(c) Skills

Programming	CUDA, HIP, C++, C, Python, Rust
Web	HTML, CSS, LESS
Language	English, French, Haitian-Creole
Database	SQL, MangoDB

(d) Publications

1. M. DuPont, C. Shen, and N. A. Murphy. Comparative Analysis of the Solar Wind: Modeling Charge State Distributions in the Heliosphere. *arXiv e-prints*, page arXiv:2012.12297, Dec. 2020.
2. M. DuPont and J. W. Murphy. Fundamental physical and resource requirements for a Martian magnetic shield. *International Journal of Astrobiology*, 20(3):215–222, June 2021.
3. M. DuPont, A. MacFadyen, and J. Zrake. Ellipsars: Ring-like Explosions from Flattened Stars. ApJL, 931(2):L16, June 2022.
4. M. DuPont, A. MacFadyen, and R. Sari. On the Theory of Ring Afterglows. ApJ, 957(1):29, Nov. 2023.
5. M. DuPont. SIMBI: 3D relativistic gas dynamics code. Astrophysics Source Code Library, record ascl:2308.003, Aug. 2023.
6. M. DuPont and A. MacFadyen. Stars Bisected by Relativistic Blades. ApJL, 959(2):L23, Dec. 2023.
7. M. DuPont, A. MacFadyen, and S. E. de Mink. Explosions in Roche-lobe Distorted Stars: Relativistic Bullets in Binaries. *ApJ*, 964(1):56, Mar. 2024.

8. M. DuPont, A. Gruzinov, and A. MacFadyen. Strong Bow Shocks: Turbulence and an Exact Self-similar Asymptotic. ApJ, 971(1):34, Aug. 2024.
9. M. DuPont and A. MacFadyen. Did Binary Neutron Star Merger GW170817 Leave Behind a Long-lived Neutron Star? ApJL, 971(1):L24, Aug. 2024.
10. M. DuPont and E. Quataert. Self-limited Accretion onto Embedded Binaries in a Uniform Medium. ApJ, 1004(2):196, June 2026.
11. M. DuPont. A Similarity Theorem and Its Breakdown in Atomic Black Hole Accretion. *arXiv e-prints*, page arXiv:2608.28587, Aug. 2026.

(e) Awards & Honors

HEAD Dissertation Prize	HEAD Executive Committee	2025
Lyman Spitzer Jr. Fellowship	Princeton University	2023
FFPS Fellowship	Princeton University	2023
SCEECS + KIPAC Fellowship (declined)	Stanford University	2023
Burke Fellowship (declined)	Caltech	2023
Kavli Summer Program in Astrophysics	University of California, Santa Cruz	2023
James Arthur Graduate Associate Fellowship	New York University	2023
Outstanding Graduate Student Instructor Award	New York University	2022
KITP Graduate Fellowship	Kavli Institute for Theoretical Physics	2022
James Arthur Graduate Associate Fellowship	New York University	2021
AAS Travel Grant	American Astronomical Society	2017
Silver Garland in Mathematics	The Ledger Media Group	2014

(f) Programs & Committees

Simons Collaboration on Extreme Electrodynamics of Compact Sources (SCEECS)	2024
National Society of Black Physicists (NSBP)	2020
American Astronomical Society (AAS)	2017

(g) Invited Presentations

1. M. DuPont. "Death Stars: Ring-explosions from flattened stars"', May 6, 2022. Caltech: Theoretical AstroPhysics Including Relativity (TAPIR).
2. M. DuPont. "Death Rays: Relativistic Beams from Roche Lobe Filling Stars"', Aug 5, 2023. Max-Planck-Institut für Astrophysik (MPA).
3. M. DuPont. "Death Stars: Discerning Astrophysical Transients From Non-Conventional Explosion Geometries"', Sep 21, 2023. Flatiron Institute: Center for Computational Astrophysics (CCA).
4. M. DuPont. "Astrophysical implications of non-conventional explosion geometries"', Nov 9 2023. Institute for Advanced Study (IAS).
5. M. DuPont. "The Life and Death of Stars"', Feb 7 2024. Bookclub Bar (NYC).
6. M. DuPont. "Explosions in Roche-libe Distorted Stars: Relativistic Bullets in Binaries"', May 8 2024. Purdue University.
7. M. DuPont. "Toroidal Explosions"', Oct 7, 2024. University of California Berkeley: Theoretical Astrophysics Center (TAC).
8. M. DuPont. "The role of asymptotic methods in astrophysics"', Feb 27, 2025. Kyoto University.

9. M. DuPont. "Non-convential Mechanisms Behind Some Astrophysical Transients"', April 29, 2025. Dartmouth University.
10. M. DuPont. "Non-convential Mechanisms Behind Some Astrophysical Transients"', May 12, 2025. Niels Bohr Institute.
11. M. DuPont. "Did Binary Neutron Star Merge GW170817 Leave Behind a Neutron Star?"', August 24, 2025. Lijiang, China.
12. M. DuPont. "Toroidal Explosions"', Sep 05, 2025. Hangzhou, China (Zhejiang University).
13. M. DuPont. "Dissertation Prize Talk"', October 13, 2025. St. Louis, Missouri (HEAD22 Meeting).
14. M. DuPont. "Binary Bondi Accretion", February 20, 2026. Kyoto, Japan (YITP).
15. M. DuPont. "Binary Bondi Accretion"', March 4, 2026. Kobe, Japan (RIKENS iTHEMS).
16. M. DuPont. "Quasi-Spherical Accretion and My Journey Towards Astronomy"', August 6th, 2026. Santa Cruz, California (Lamat Institute, UCSC).
17. M. DuPont. "Toroidal Sky Maps", August 13, 2026. West Lafayette, Indiana (Purdue University).

(h) Invited Lectures

1. M. DuPont. "An Introduction to Numerical Hydrodynamics", March 14 2024. Flatiron Institute: Center for Computational Astrophysics (CCA).
2. M. DuPont. "Introduction to Numerical Magnetohydrodynamics", April 23 2025. Columbia University.

(i) Synergistic Activities

1. Popular Media
   - LinkNYC (Apr, 2023) — Cosmic Curiosity, My research was featured throughout the entire city of New York through interactive visual kiosks showcasing images from my high-resolution simulations of exploding stars.
   - AAS NOVA (Mar, 2024) — How to Slice a Star, My research on relativistic blades was highlighted by the enthusiast-centered arm of the American Astronomical Society.
2. Advising
   - Planette Analytics Inc. – Consulting Physicist and Scientific Advisor
     - Provide scientific guidance for a NASA-funded climate-forecasting proof of concept, developing and validating predictive reduced-order models using dynamical systems, information theory, symmetry, and topology.
3. Teaching
   - FSU (Aug, 2018 - Dec, 2018) — Physics Problem Solving
     - Developed a curriculum that was focused around helping students build physics intuition by means of order-of-magnitude focused exercise
   - NYU (Aug, 2021 - Dec, 2021) — Computational Physics
     - Taught with a focus on signal processing, dynamics, and optimization techniques.
   - Princeton Prison Teaching Initiative (Jun, 2026 - Aug, 2026) — Intro to Astronomy
     - Teaching Intro to Astronomy to formally incarcerated individuals.
   - Princeton Prison Teaching Initiative (Aug, 2026 - Present) — Basic College Mathematics
     - Instructor for Basic College Mathematics, Northern State Prison

4. Mentor
   - FIRST Lego Robotics (Aug, 2013 - Oct, 2013)
     - Teach kids simple coding methods utilizing Arduino boards coupled with the `Scratch` build block programming scheme. These robots were then used in competition with other in-state institutions.
   - STEM Scholarbotics (Aug, 2013)
     - Help students virtually perform surgery using digital Davinci arm simulation programs to provide hands-on experience with cutting edge technology.